

 UNIVERSITI KEBANGSAAN MALAYSIA <i>National University of Malaysia</i>		ASASIpintar Pusat PERMATA@Pintar Negara	
LECTURERS/ INSTRUCTORS:		COURSE OUTLINE DETAILS PNAP0113: BIOLOGY I SEMESTER I, 2026/2027	
Week	Date	Topic	Activity
1	29 Jun –3 July	1. Biochemistry • Atoms, molecules, ions and bonds • Water	Lecture (4 hrs)-IZ
2	6 – 10 July	• carbohydrates, lipids, protein and nucleic acid • Enzymes and metabolism	Lecture (4 hrs)-IZ Tutorial (1hr)
3	13 – 17 July	2. The Cell • Cell theory and methods of studying cells • Structure and function of the cell • Transport into and out of the cell • Cell communication	Lecture (4hrs)-SEL <u>Quiz 1 (IZ)</u> Tutorial : (1hr)
4	20– 24 July	3. Cell respiration • Glycolysis, The Krebs cycle • Oxidative phosphorylation • Chemiosmosis • Anaerobic respiration • Lactic acid fermentation	Lecture (4hrs) - SEL Tutorial: (1hr)

5	27 – 31 July	4. Photosynthesis • Light dependent reaction • Noncyclic and cyclic photophosphorylation	Lecture (4hrs) -IZ Practical 1 (2hrs) Microscopy
6	3 – 7 Aug	• Light independent reaction • Photorespiration • C4 and CAM plants	Lecture (4hrs)-IZ Tutorial (1hr)
7	10– 14 Aug	5. Plant Physiology • Plant tissue • Transport in plants	Lecture (4hrs)- <u>MF</u> Quiz 2 (SEL)
16-23 Aug 2026 MID SEM BREAK			
24-28 Aug 2026 MID SEMESTER I EXAM			
8	*31Aug– 4 Sept (*31 Aug Mon Holiday)	• Plant responses to stimuli	Lecture (4hrs) - MF Tutorial (1hr)
9	7 -11 Sept	6. Animal Physiology • Temperature regulation • Osmoregulation	Lecture (4hrs)- NN Tutorial (1hr)
10	*14-18 Sept (*16 Sept Hari Malaysia)	• Excretion • Nervous system and muscles	Lecture (4hrs)- NN Tutorial (1hr)
11	21- 25 Sept	• Digestion • Gas exchange	Lecture (4hrs)- SEL Practical 2 (2hrs) Diffusion & Osmosis
12	28 Sept- 2 Oct	• Circulation • Chemical signal	Lecture (4hrs)-SEL Tutorial (1hr)

13	5 – 9 Oct	7. Human Immune System • Non Specific and Specific Defense mechanisms • First line of defense system	Lecture (4hrs)- IZ Quiz 3 (MF) Tutorial (1hr)
14	12– 16 Oct	• Second line of defense system • Third line of defense system • Cell- mediated response • Humoral response	Lecture (4hrs)-IZ Tutorial (1hr)
15	19– 23 Oct	8. Animal Reproduction and development Asexual	Lecture (4hrs) - NN Practical 3: (2hrs) (Chromatography& Photosynthesis)
16	26– 30 Oct	• Sexual reproduction • Embryonic development	Lecture (4hrs) - MF Tutorial (1hr)
17	2 - 6 Nov	*Field work/Visit- Anatomy Museum (HCTM)	
9 - 15 NOV 2026: REVISION WEEK			
16 – 20 NOV 2026			
FINAL SEMESTER EXAMINATION			
23 NOV – 6 DEC 2026 SEM 1 HOLIDAY			